

The empirical recurrence for OEIS A230172, recovered from its tail

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Abstract

OEIS A230172 records “Empirical recurrence of order 81” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 102$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 83$. Its last coefficient is $c_{81} = 105136128 \neq 0$, so no further tail satisfies anything shorter; any order-81 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A230172 is “Number of $n \times 4$ 0..2 arrays $x(i,j)$ with each element horizontally, vertically, diagonally or antidiagonally next to at least one element with value $2-x(i,j)$.”. It has offset 1 and begins

9, 1407, 78849, 4722977, 284444079, 17072965263, 1027868078181, 61861526624839, ...

The entry states:

Empirical recurrence of order 81 (see link above).

As of the “Last modified” line on the live entry (Feb 22 2025 14:36:10, revision 9) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 102$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 102$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 81: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 81.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 204$ exact terms from $n = 2$ on, it returns order 81 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{81} c_i a(n-i),$$

c_1	45	c_{22}	12739133634121162	c_{43}	−283810539918170080887	c_{64}	−125985151561
c_2	325	c_{23}	88553575264390869	c_{44}	190214851734842081492	c_{65}	818319652104
c_3	29395	c_{24}	−217235871173578281	c_{45}	153248555076721933546	c_{66}	−40738929218
c_4	315851	c_{25}	−104298131103939478	c_{46}	−132893903731263869485	c_{67}	293021791607
c_5	2682434	c_{26}	1346433266616575266	c_{47}	−157607528398073681115	c_{68}	−16252948210
c_6	−3189825	c_{27}	928493080700891154	c_{48}	292117805440459103888	c_{69}	247235439002
c_7	−173580099	c_{28}	−5557219223068290846	c_{49}	−1176467478872954794	c_{70}	−9577549547
c_8	−1151841106	c_{29}	1393355728816767875	c_{50}	−183424980933194280627	c_{71}	−3221356897
c_9	−2884176202	c_{30}	4532979992342362727	c_{51}	−51786536574633723864	c_{72}	−2875762549
c_{10}	3457744885	c_{31}	−5385235524551746231	c_{52}	317156841512542133503	c_{73}	−1271536213
c_{11}	98717057498	c_{32}	−13281328075671168834	c_{53}	−101326195127018938203	c_{74}	13250790142
c_{12}	777925061314	c_{33}	10578422531905759439	c_{54}	−149760331314681264624	c_{75}	4524197097
c_{13}	4996378737056	c_{34}	−16041601780094317484	c_{55}	79806022829036875850	c_{76}	−207459290
c_{14}	−247748892022	c_{35}	1927835826742001447	c_{56}	56779629639663739137	c_{77}	103339239
c_{15}	−44794699770516	c_{36}	148430986230921327729	c_{57}	−26846969543111787913	c_{78}	−93370318
c_{16}	−192438747789538	c_{37}	−96142787466400090982	c_{58}	−12877039366693496113	c_{79}	58274196
c_{17}	315626583036945	c_{38}	−1865959788880613269	c_{59}	3520320579456404523	c_{80}	18678988
c_{18}	1125839323389627	c_{39}	−17238121340090083219	c_{60}	3613381517294190290	c_{81}	10513612
c_{19}	4881221022881715	c_{40}	−166974171127414928145	c_{61}	95986526364845868		
c_{20}	−9074707732648533	c_{41}	370329958812277188965	c_{62}	−1363651871488635697		
c_{21}	−23334952797305049	c_{42}	−203391177705643558610	c_{63}	−527339339980083071		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 83$, and it is the recurrence OEIS A230172 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{81} - c_1x^{81-1} - \dots - c_{81}$. Its constant term is $-c_{81} = -105136128$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 81 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 81, so $\deg q = 81$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 81, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 81, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 105136128.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-81 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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